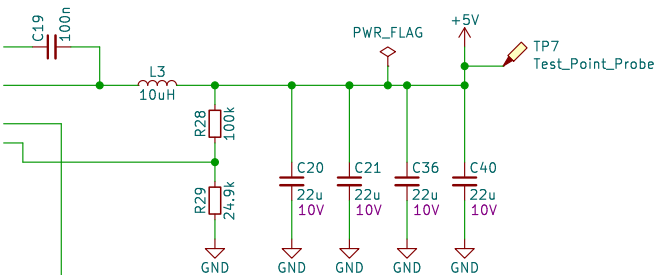
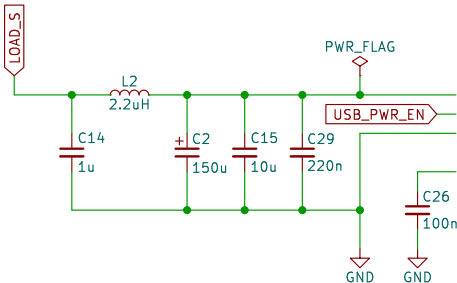


Battery to 5V (SMPS)

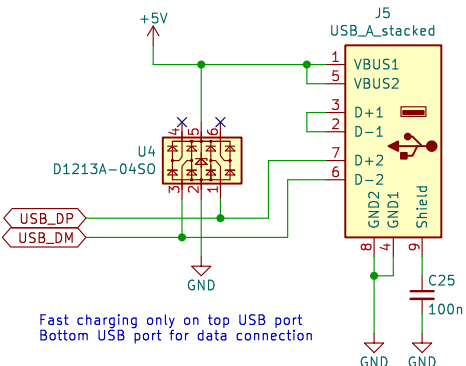
Output voltage: $0.925 \times (1 + 45.3/10) = 5.12 \text{ V}$

Layout for 2A output current
(1x up to 1.5A fast charging USB port, one normal 500 mA port)



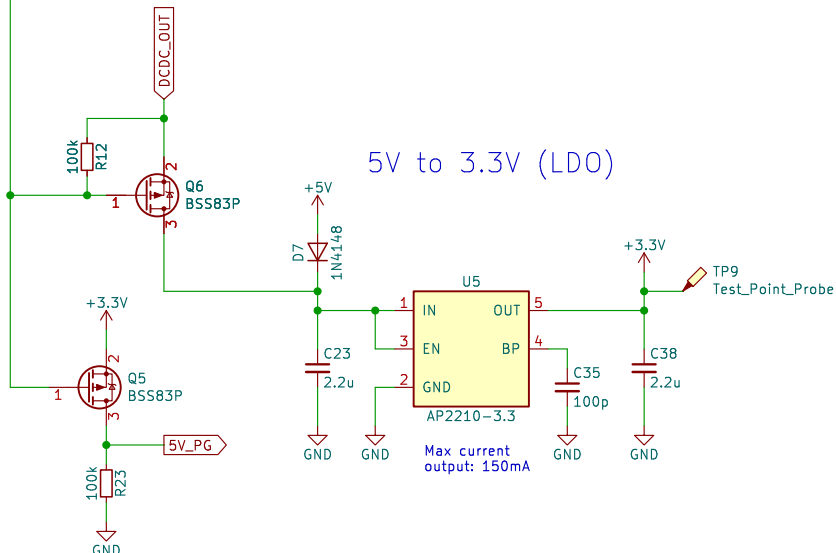
Fall-back power supply
from 12V rail if 5V is
not good enough

USB connector and power supply



Fast charging only on top USB port
Bottom USB port for data connection

5V to 3.3V (LDO)



Licensed under CERN-OHL-W version 2

Author: Martin Jäger

Licensed under CERN-OHL-W version 2

Copyright © 2019 Libre Solar Technologies GmbH

Sheet: /

File: working.kicad_sch

Title: MPPT 1210 HUS

Size: A4	Date: 2021-05-02
----------	------------------

KiCad E.D.A. eeschema 7.0.6

Rev: 0.7.1

Id: 1/1

If not stated otherwise, all MLCC 50V X7R